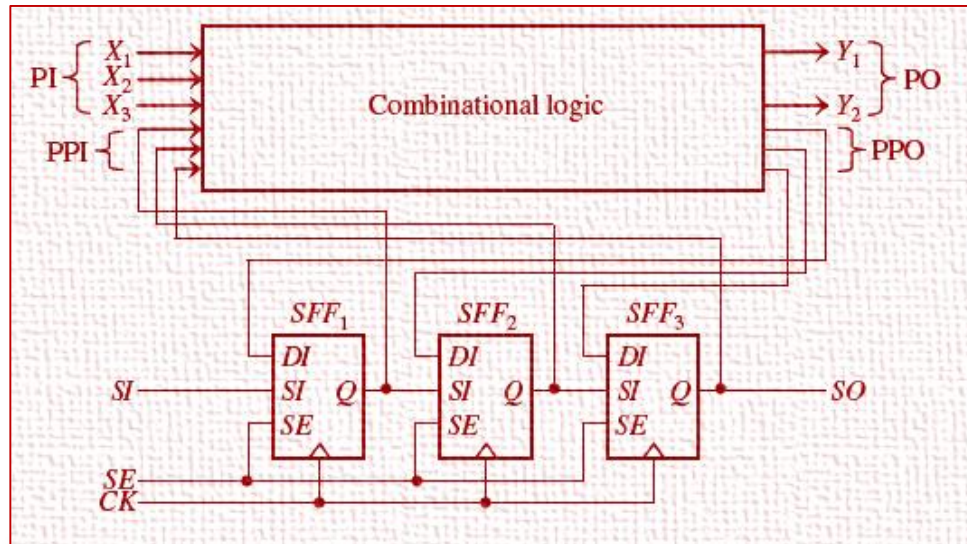
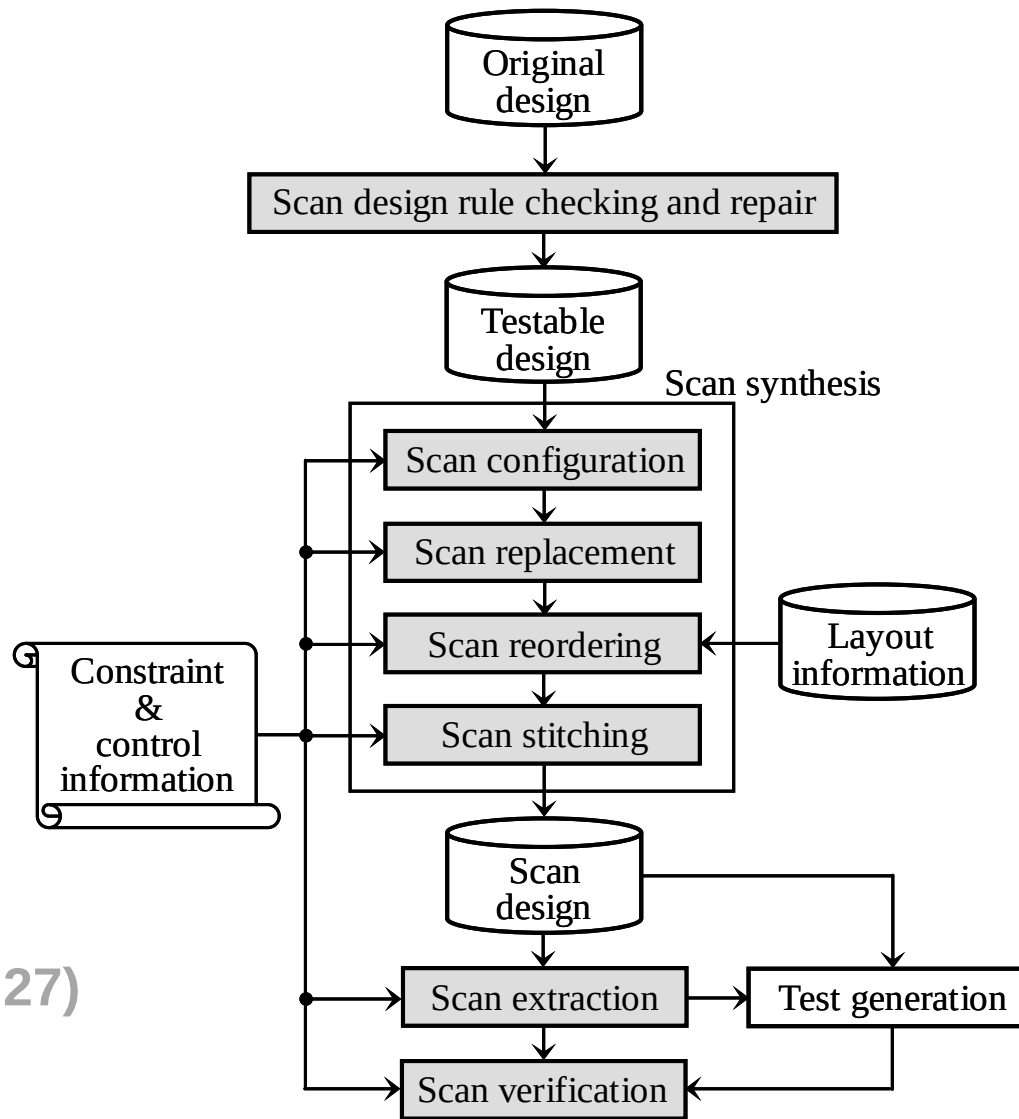


DFT - Part 1

- Introduction
- Internal Scan
 - ◆ FF-based
 - ◆ Latch-based
- **Scan Design Flow**
- Issues and Solutions
- Conclusion



Scan Design Flow



(WWW Fig. 2.27)

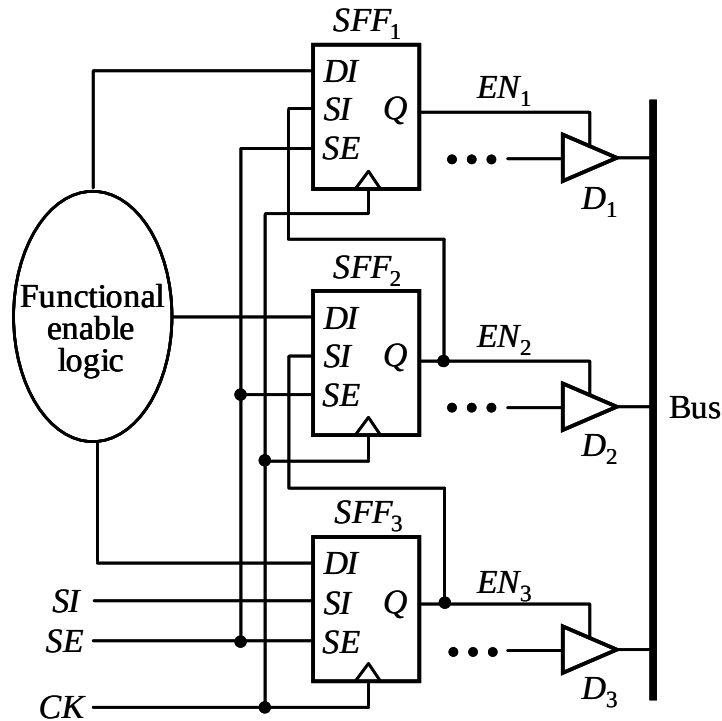
Scan Design Rule Checking

- **DRC:** Make sure designs obey rules so that scan chains can be inserted
 - ♦ Tools now supports automatic fix violations after DRC
- Typical scan design rules (WWW Table 2.7)

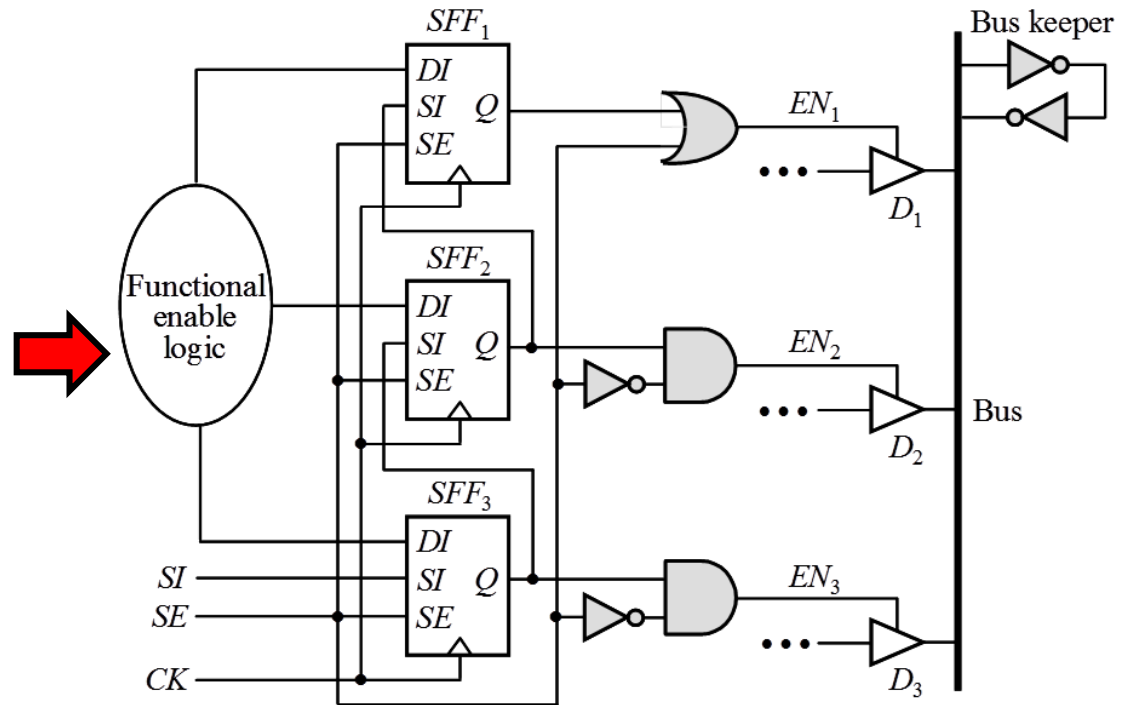
	<i>Design Style</i>	<i>Scan Design Rule</i>	<i>Recommended Solution</i>
DRC#1	Tri-state buses	Avoid during shift	Fix bus contention during shift
DRC#2	Bi-directional I/O ports	Avoid during shift	Force to input or output mode during shift
DRC#3	Gated clocks (muxed-D full-scan)	Avoid during shift	Enable clocks during shift
	Derived clocks (muxed-D full-scan)	Avoid	Bypass clocks
DRC#4	Combinational feedback loops	Avoid	Break the loops
	Asynchronous set/reset signals	Avoid	Use external pin(s)
	Clocks driving data	Avoid	Block clocks to the data portion
	Floating buses	Avoid	Add bus keepers
	Floating inputs	Not recommended	Tie to Vcc or ground
	Cross-coupled NAND/NOR gates	Not recommended	Use standard cells
	Non-scan storage elements	Not recommended for full-scan Design	Initialize to known states, bypass, or make transparent

DRC#1: Tri-State Buses

- Problem: potential bus contention during scan shift



Original design

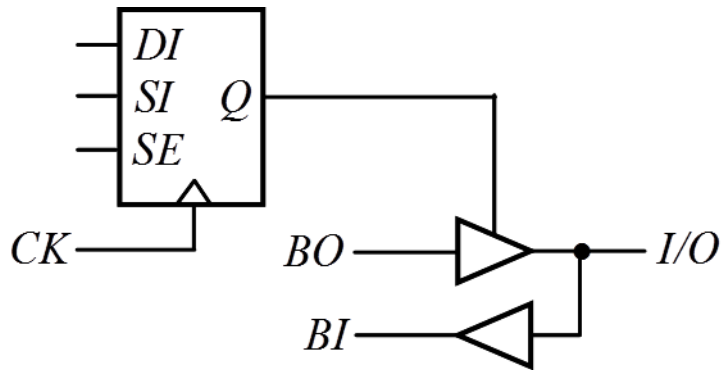


Testable design

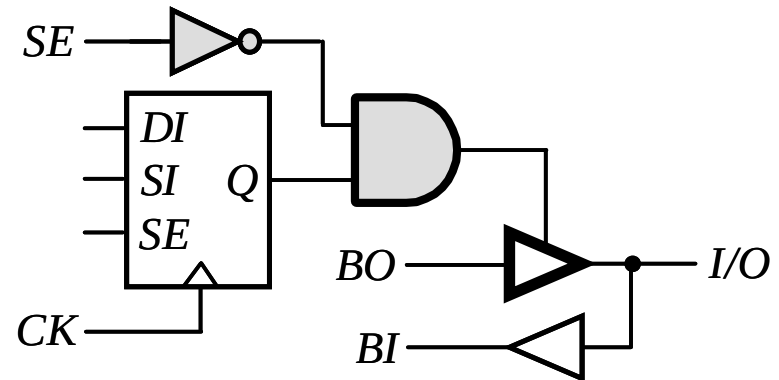
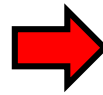
No bus contention when $SE=1$
No floating bus

DRC#2: Bidirectional I/O

- Problem: random I/O direction during scan shift



Original design



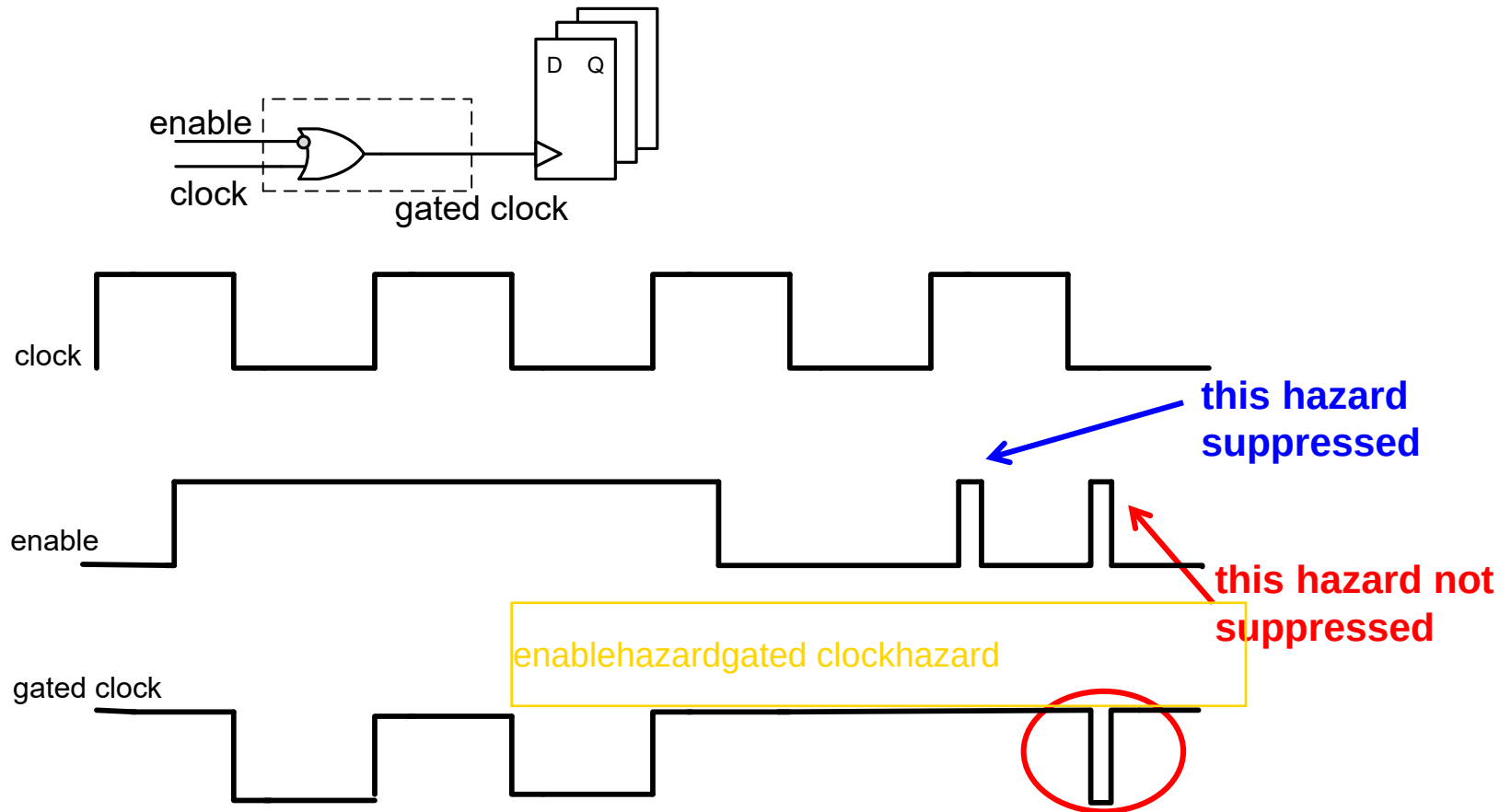
Testable design

always input pin when $SE=1$

(WWW Fig 2.22)

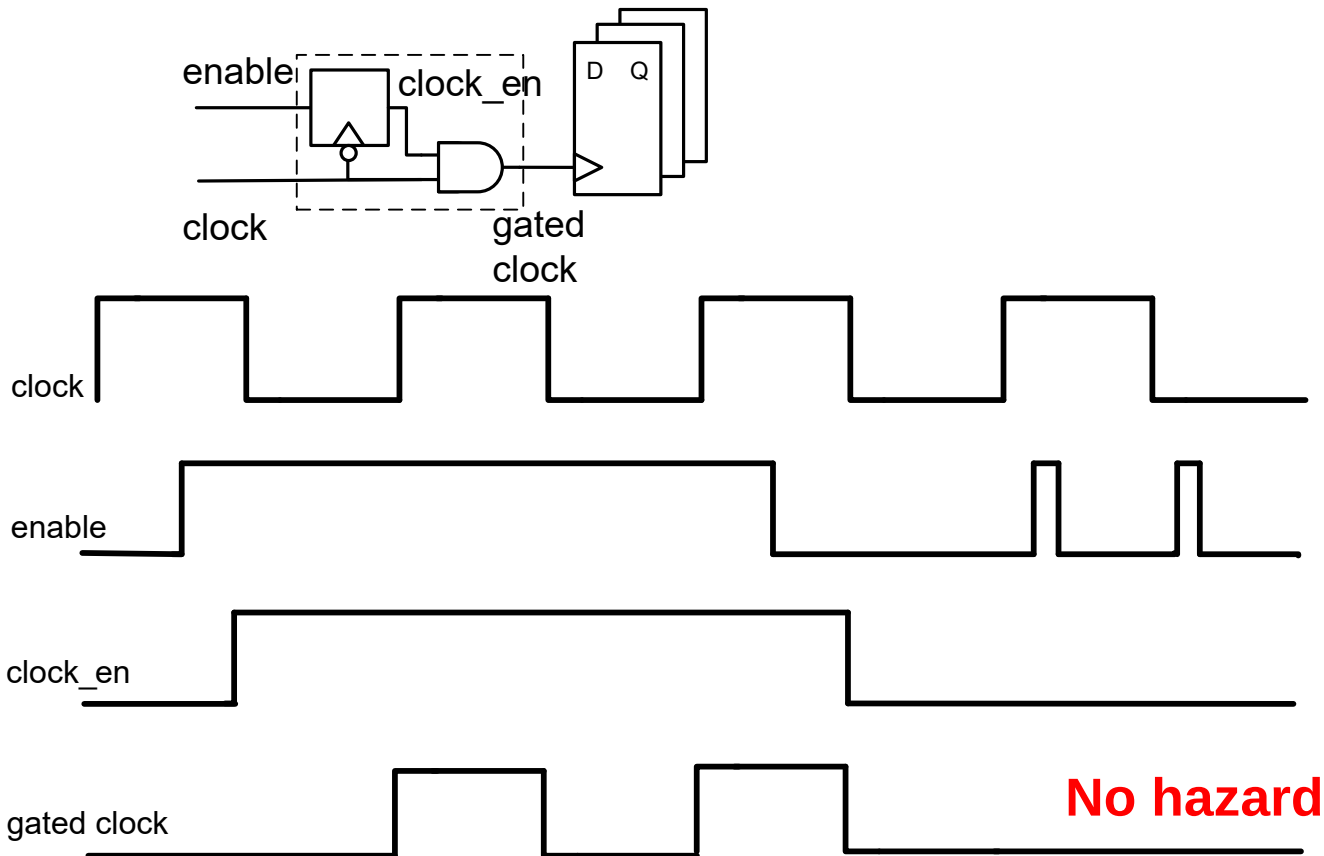
Gated Clock (1)

- Clock is gated (OFF) when enable = 0
- This is not good design. Cannot handle hazards



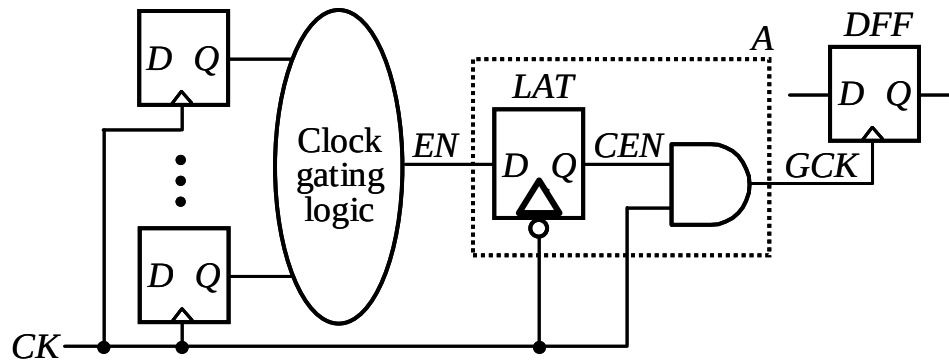
Gated Clocks (2)

- This is a better design that handle hazards
- Problem: clock is **randomly gated** during scan shift

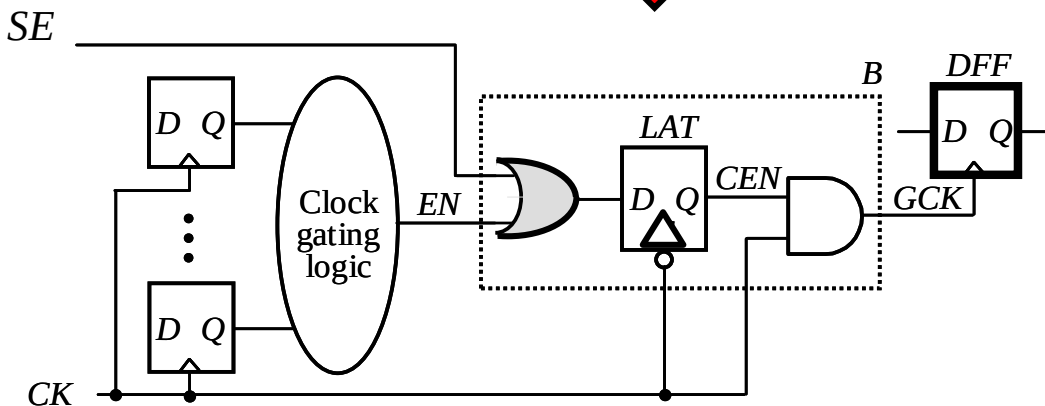


DRC#3: Gated Clock

- Problem repaired: When $SE=1$, clock always ON



Original design

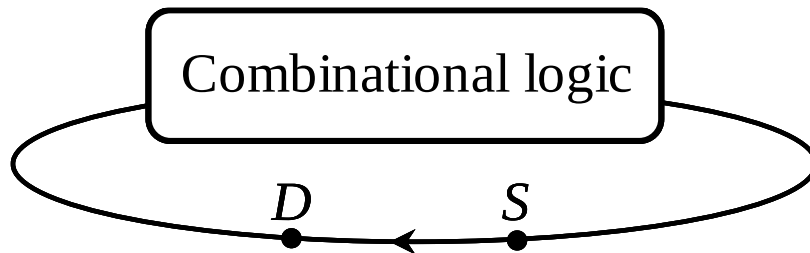


Testable design
clock always ON when $SE=1$

(WWW Fig. 2.23)

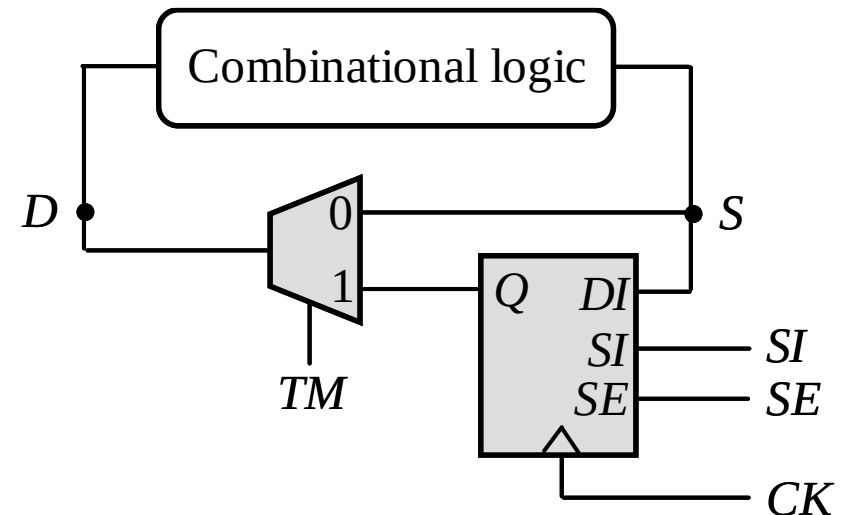
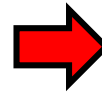
DRC#4: Combinational Feedback Loop

- Problem: Sequential behavior in test mode
 - ◆ Sequential ATPG is difficult!



(WWW Fig. 2.25)

Original design



Testable design

break loop when $TM=1$

$TM=0$, normal operation

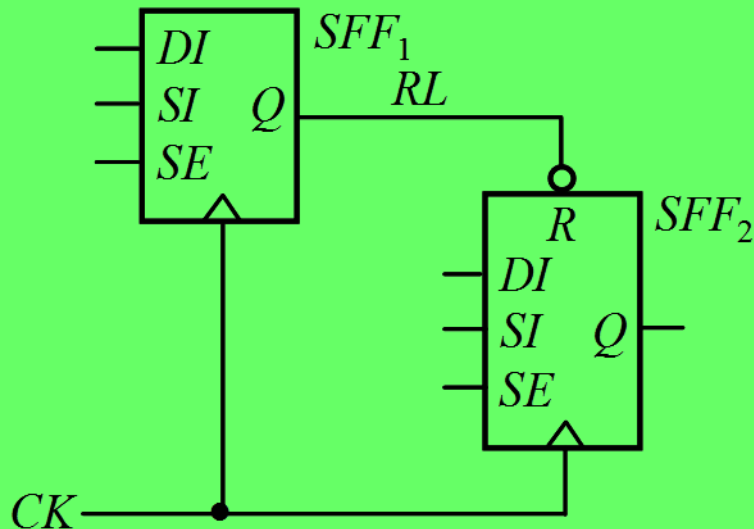
$TM=1, SE=1$: scan shift

$TM=1, SE=0$: scan capture

Quiz

Problem: SFF_2 randomly reset during scan shift

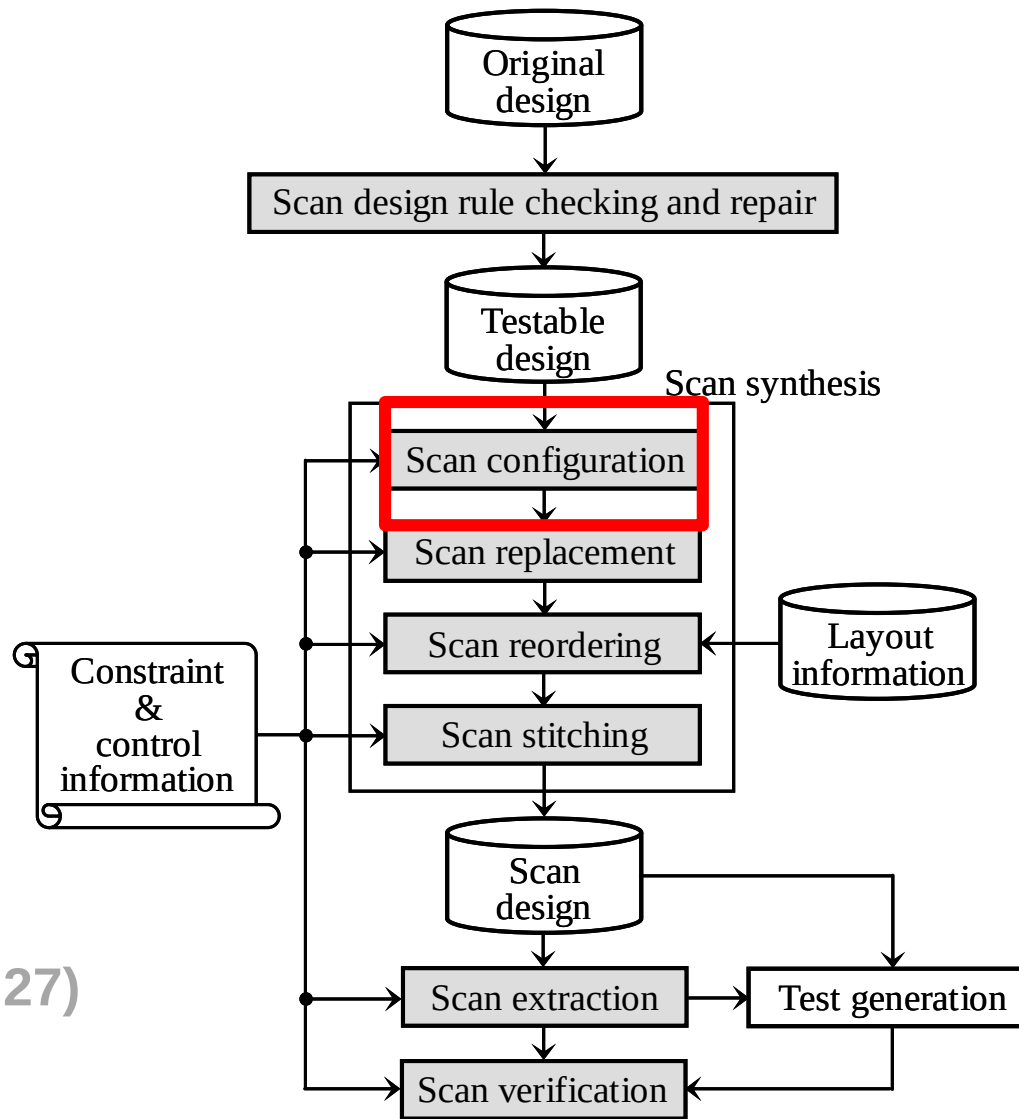
Q: Please repair this circuit so that no reset occurs when $SE=1$



Original design

Testable design

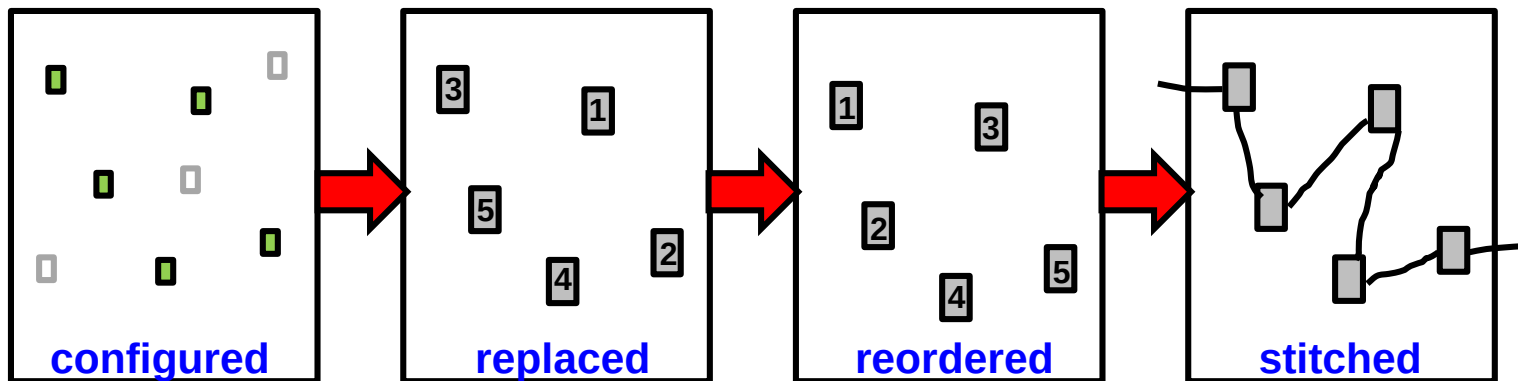
Scan Design Flow



(WWW Fig. 2.27)

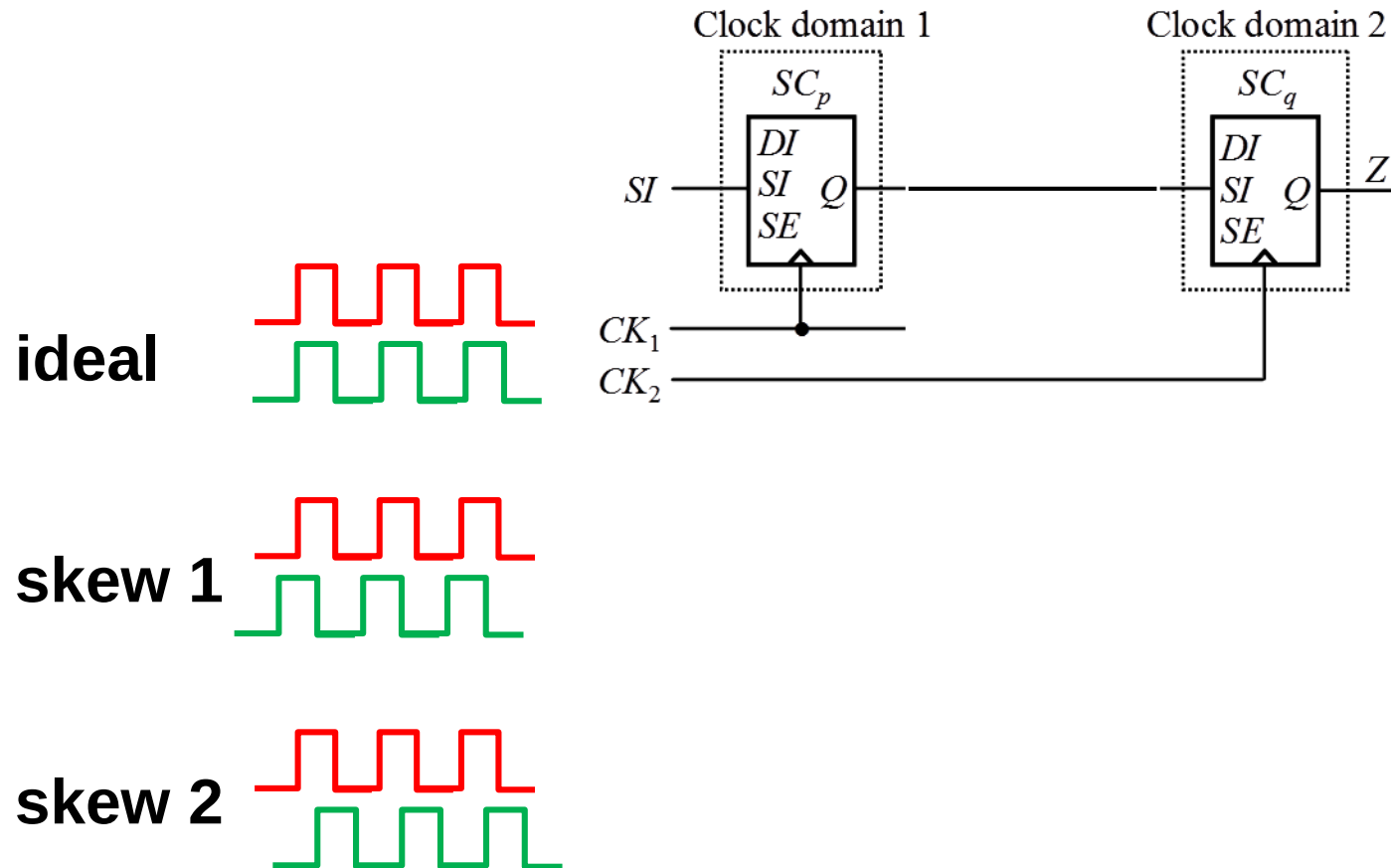
Scan Synthesis

- aka. **Scan insertion, DFT insertion**
- Four steps:
 - ♦ 1. **Scan Configuration**
 - * Decide number of scan chains, FF to exclude, ...
 - ♦ 2. **Scan Replacement**
 - * Replaces original FF/latch in testable design with their functionally-equivalent scan cells
 - ♦ 3. **Scan Reordering**
 - * Reorder scan cells to minimize routing wires
 - ♦ 4. **Scan Stitching**
 - * Stitch all scan cells together to form scan chains



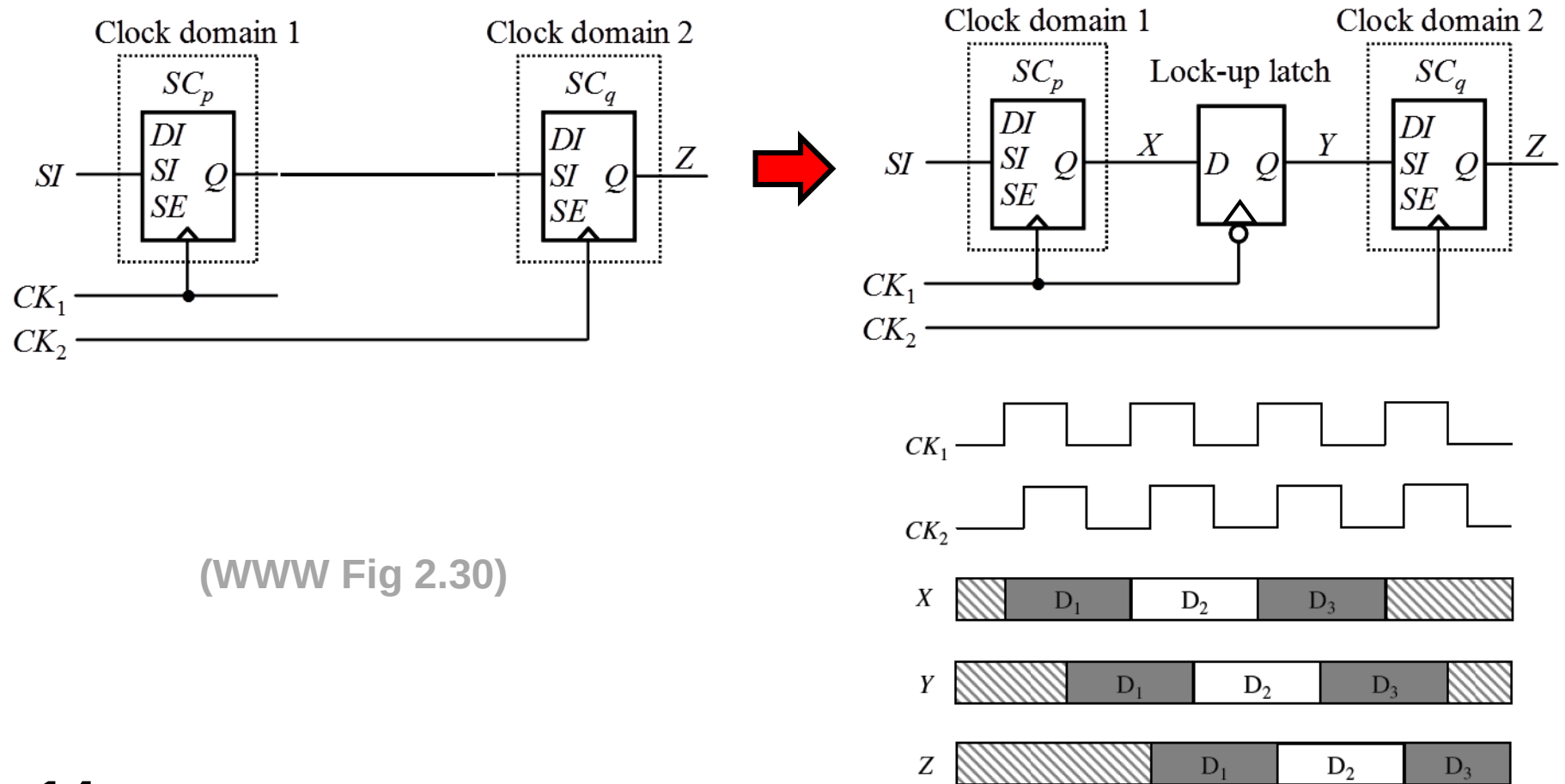
Scan Chain Cross Clock Domains

- What is wrong with this scan chain?
 - ♦ HINT: consider **clock skew** among two clock domains



Lock-up Latch

- Insert *lock-up latch* between two clock domains
 - ♦ Avoid clock skew



(WWW Fig 2.30)

Clock Grouping^{*} not in exam

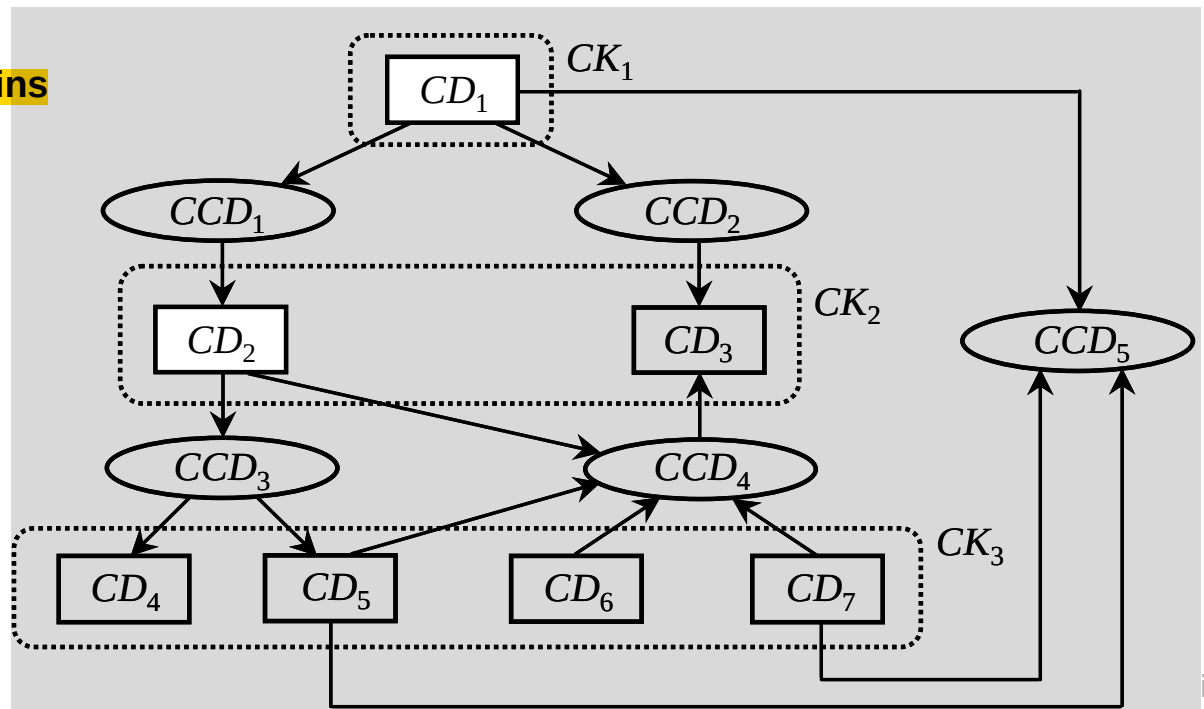
- Due to large clock skew, data from different clock domains **CANNOT** be captured together
 - ♦ CD_1 and CD_2 cannot form a scan chain.
- We can assign independent CD as one **clock group**
 - ♦ CD_2 and CD_3 has NO interaction. Can form a scan chain.
 - ♦ Totally **three** clock groups needed in this design: $CK_1 \sim CK_3$

CD: (functional) clock domains

CCD: cross clock domains

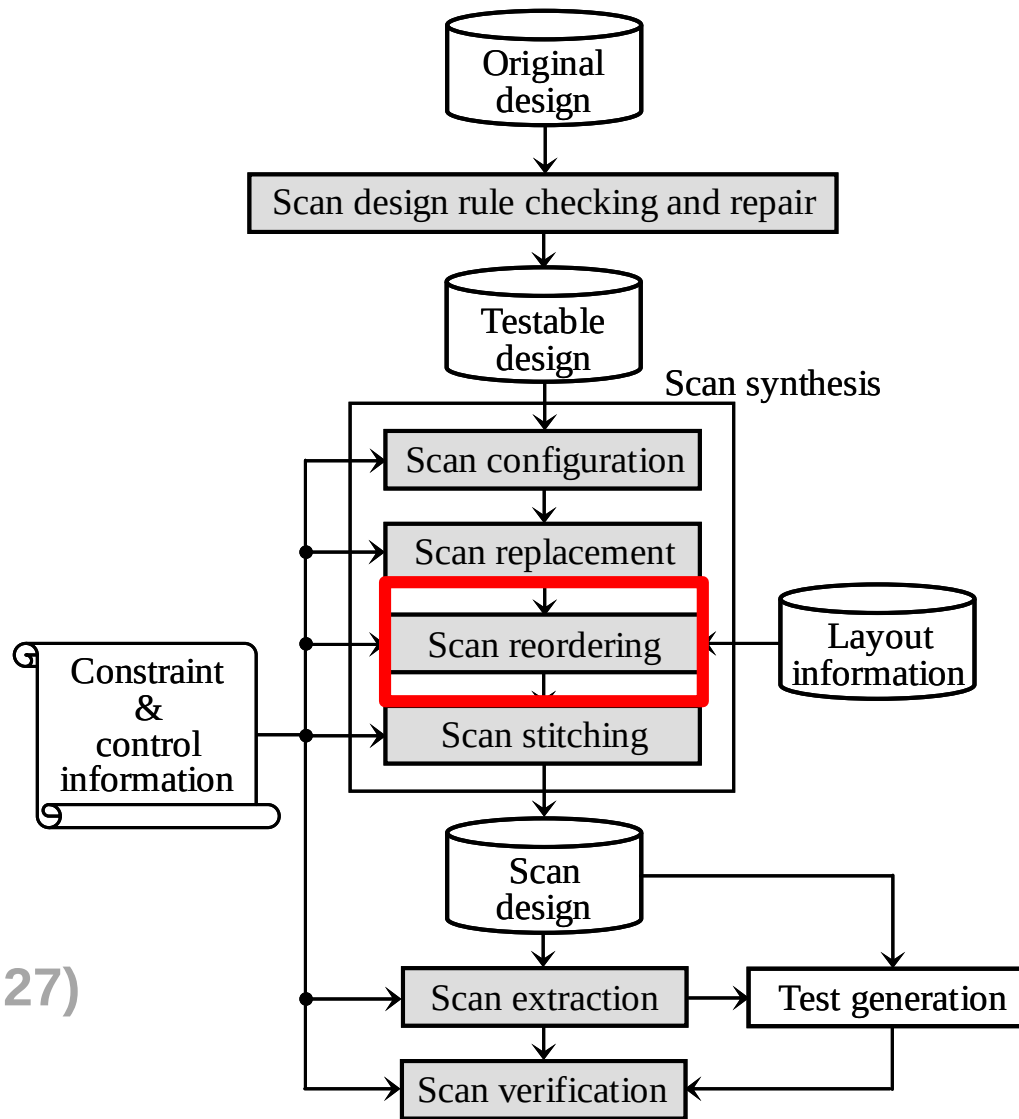
edge indicates data transfer

CK: clock groups



(WWW Fig. 2.28)

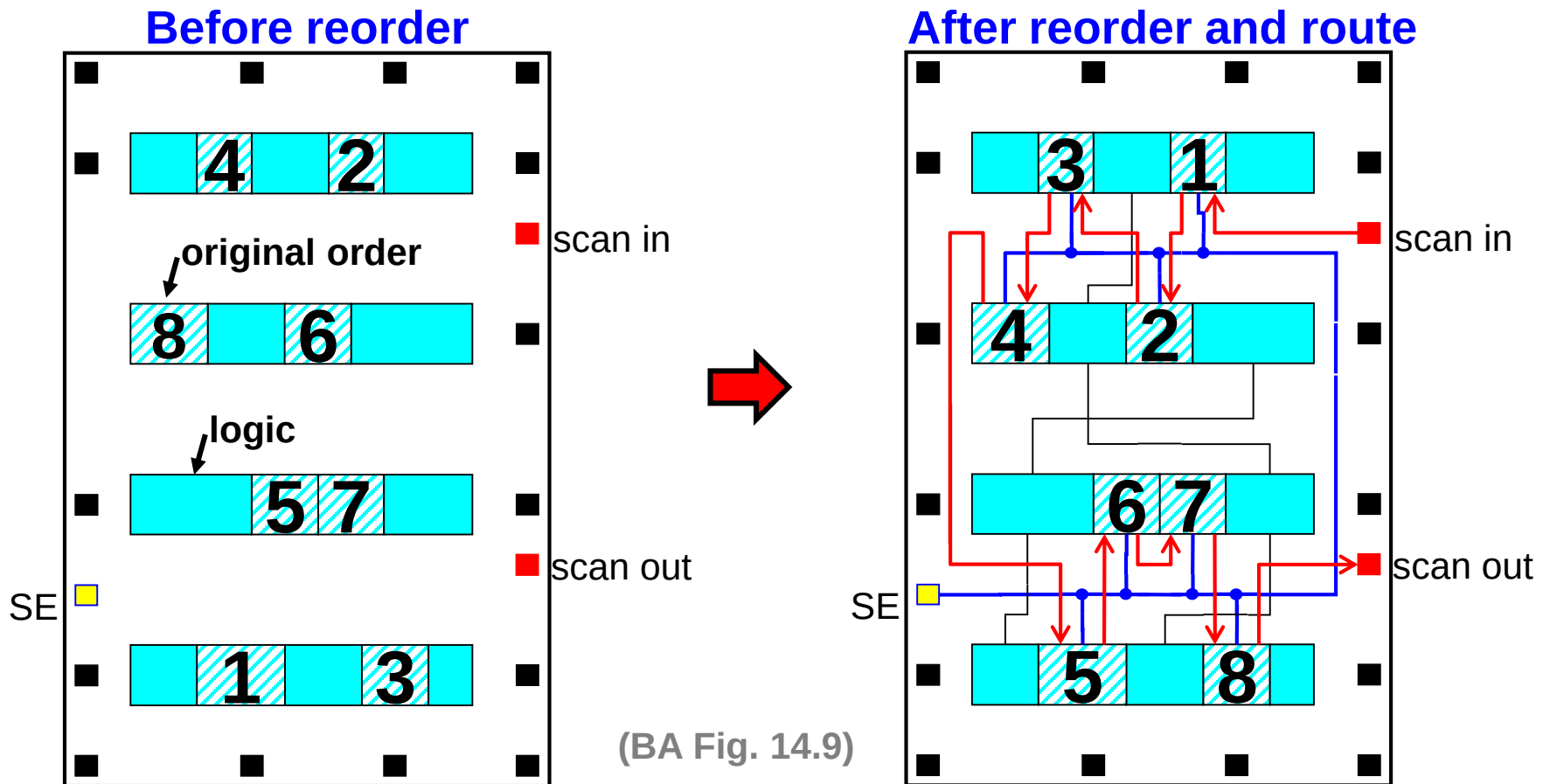
Scan Design Flow



(WWW Fig. 2.27)

Scan Chain Reordering

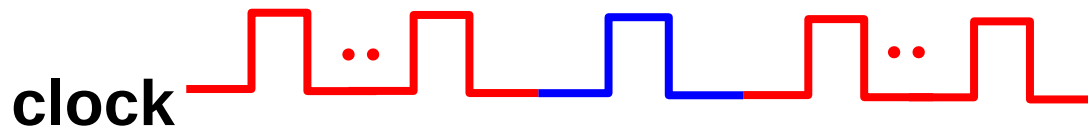
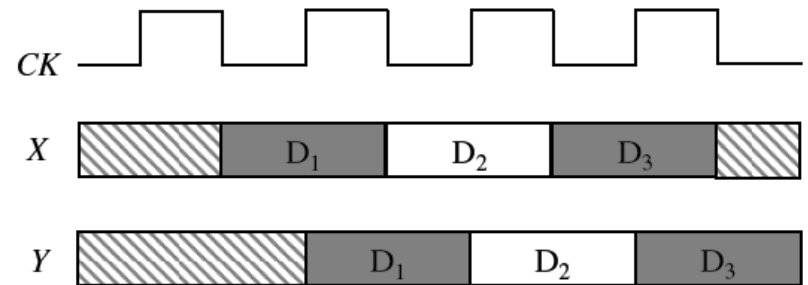
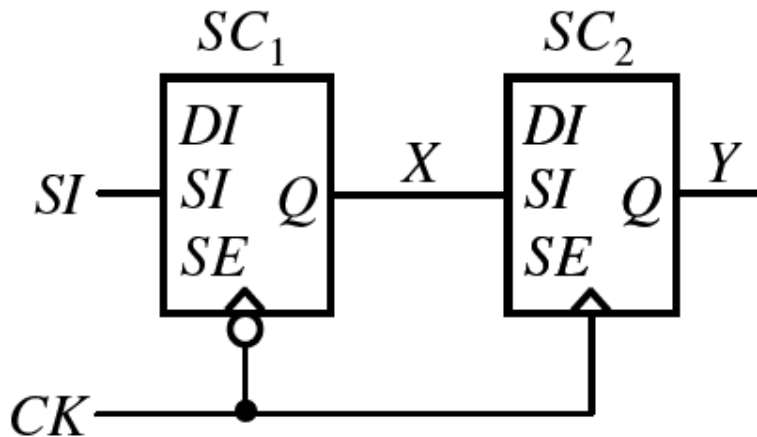
- 1. Place logic gates and FF based on functional timing
- 2. Reorder scan cells to save routing wires
- 3. Route function signals and scan chains



Mixed N/P-edge FF

- OK when **negative-edge BEFORE positive-edge** FF
- Both scan shift (SE=1) and scan capture (SE=0) are correct

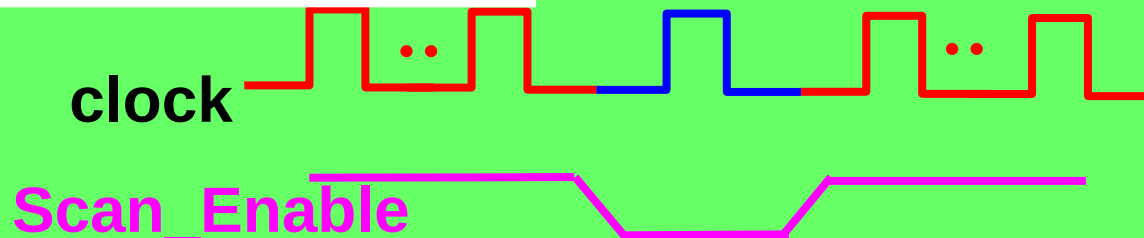
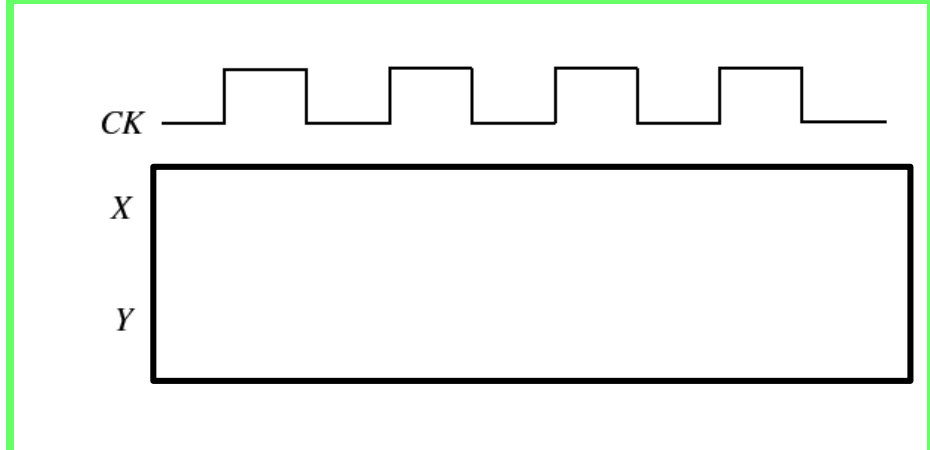
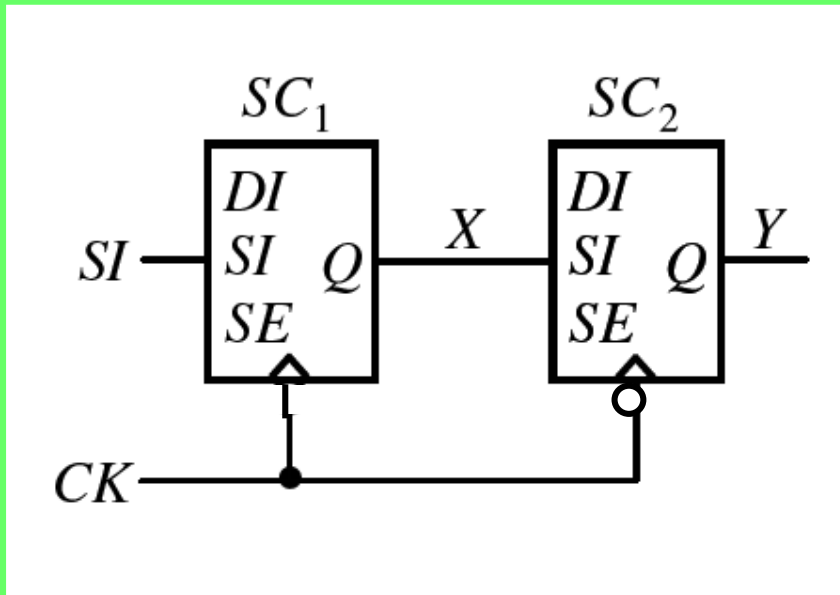
(WWW Fig 2.29)



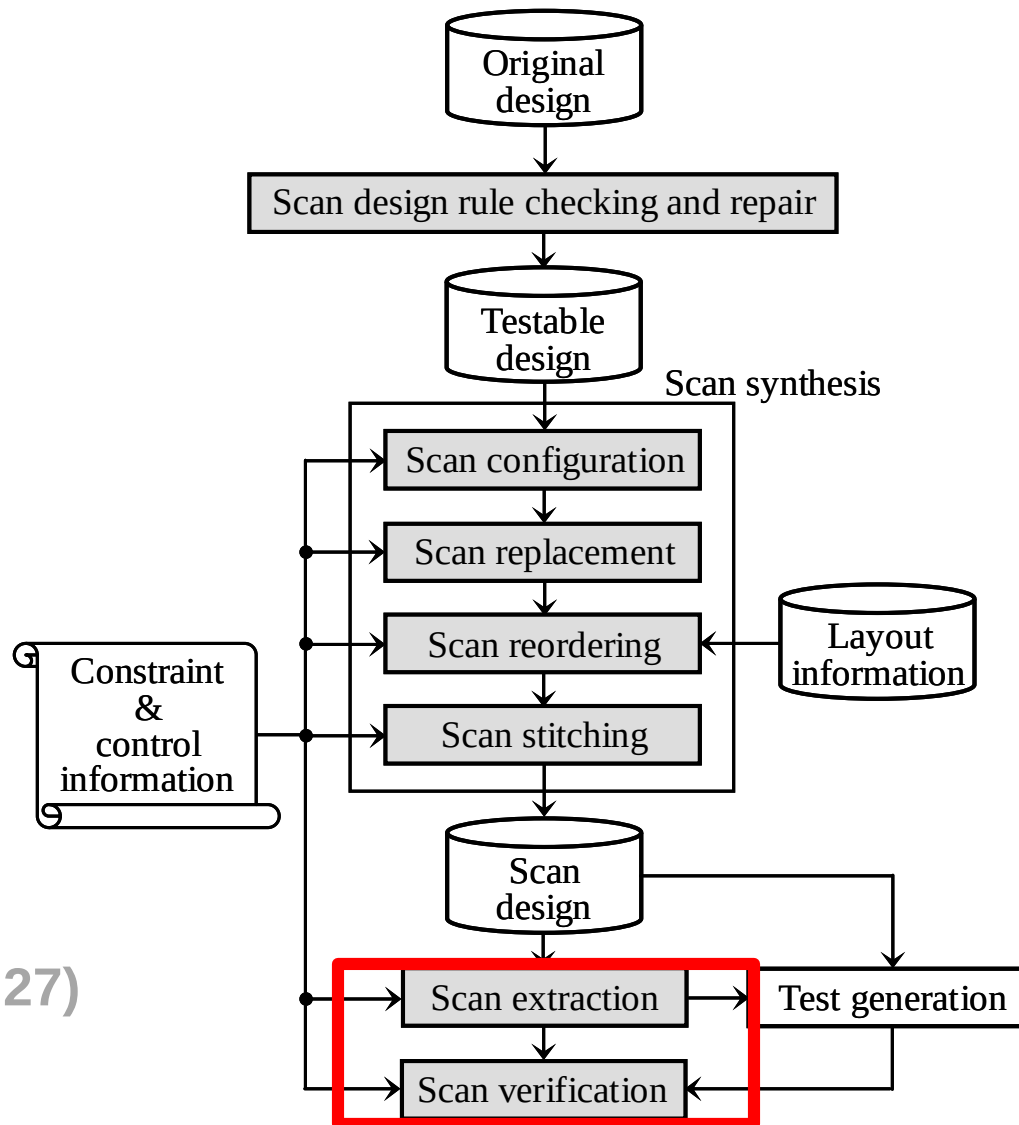
Scan_Enable

Quiz

WRONG if **positive-edge BEFORE** negative-edge FF
WHY? Hint: consider scan capture when $SE=0$



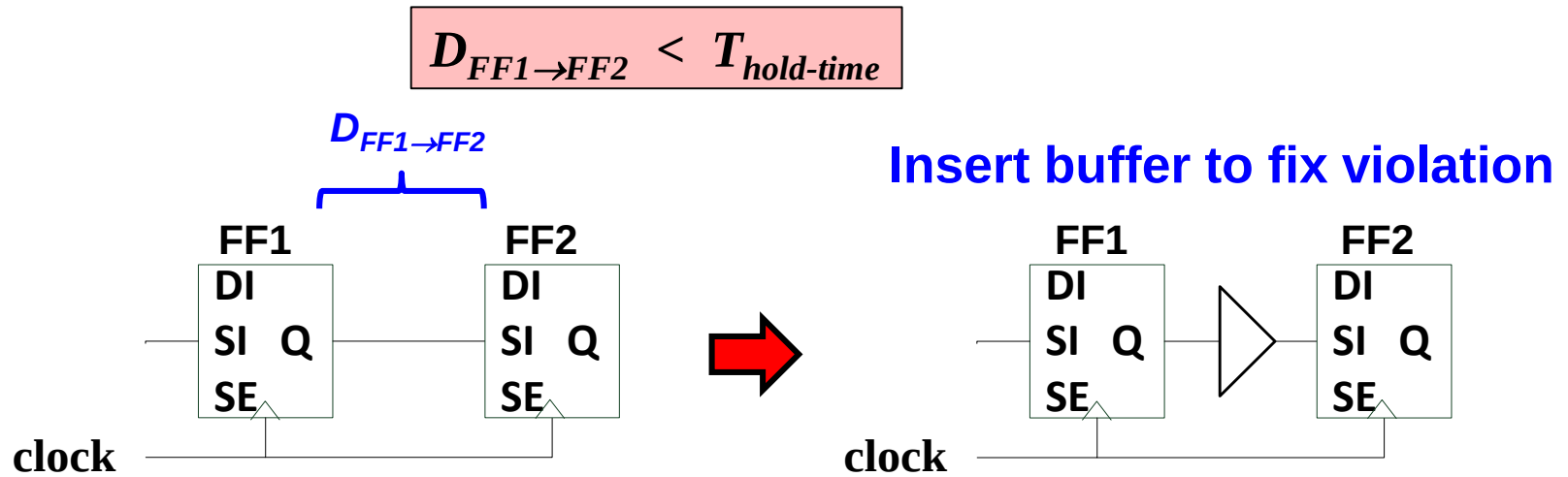
Scan Design Flow



(WWW Fig. 2.27)

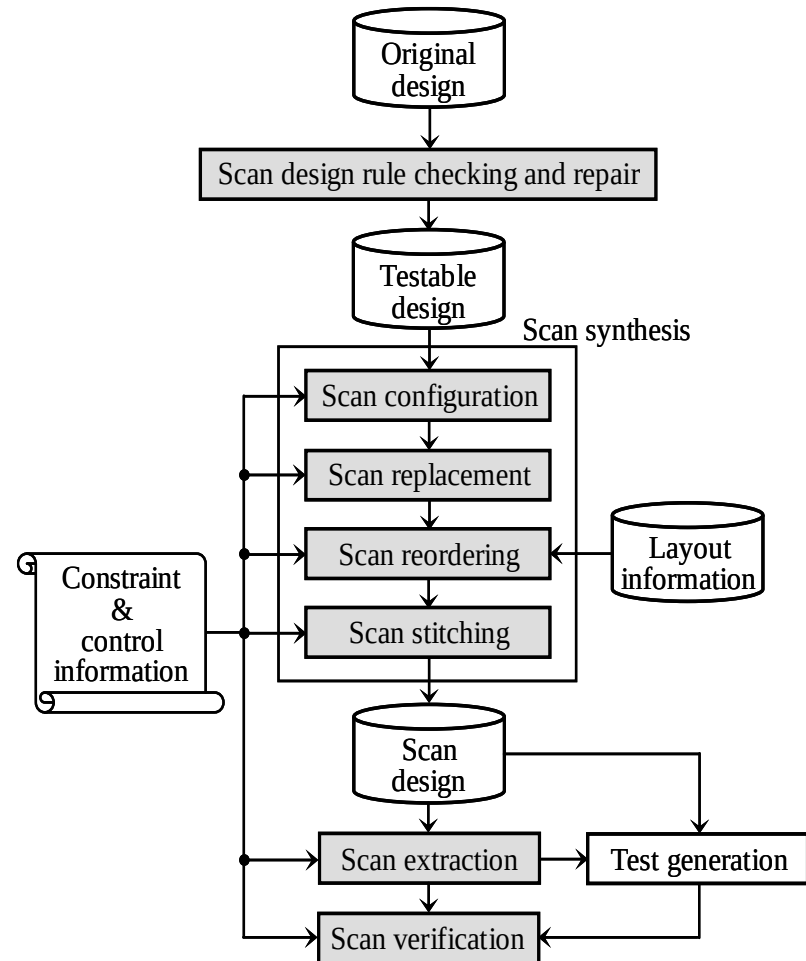
Scan Extraction and Verification

- **Scan Extraction** (*ATPG* and *Scan Synthesis* are separate tools)
 - ♦ 1. Extract order of scan cells for ATPG
 - ♦ 2. Make sure all chains are correctly connected
- **Scan Verification**
 - ♦ Make sure ATPG test patterns correctly applied
 - * both scan shift and scan capture
 - ♦ Example: timed logic simulation can catch **hold-time violation**



Summary

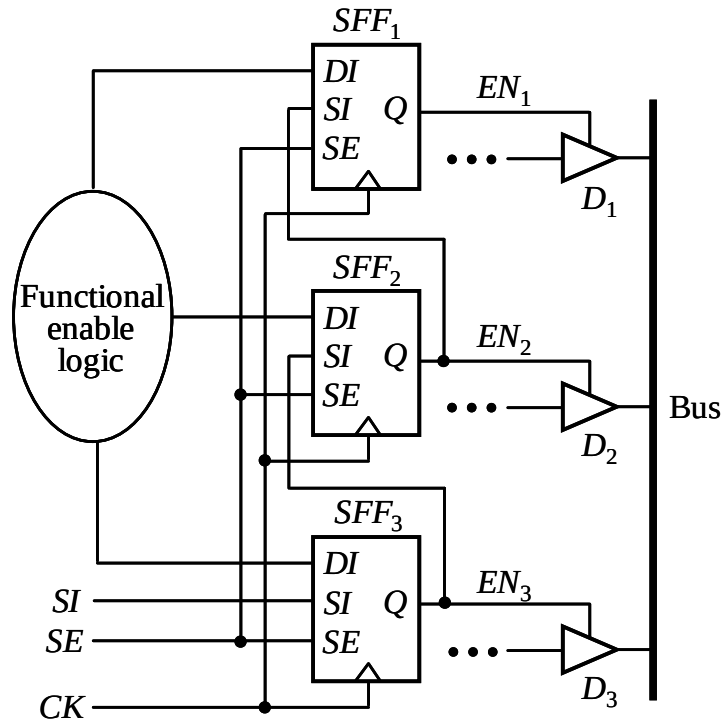
- **Scan Design Rule Checking (DRC)**
 - ◆ Tristate buses, bi-directional I/O, gated clocks, feedback loops...
 - ◆ Violations must be repaired
- **Scan Synthesis**
 - ◆ Scan configuration, replacement, reorder, stitching
 - ◆ Insert lock-up latch, buffer
- **Scan Extraction and Verification**
 - ◆ Test patterns work correctly



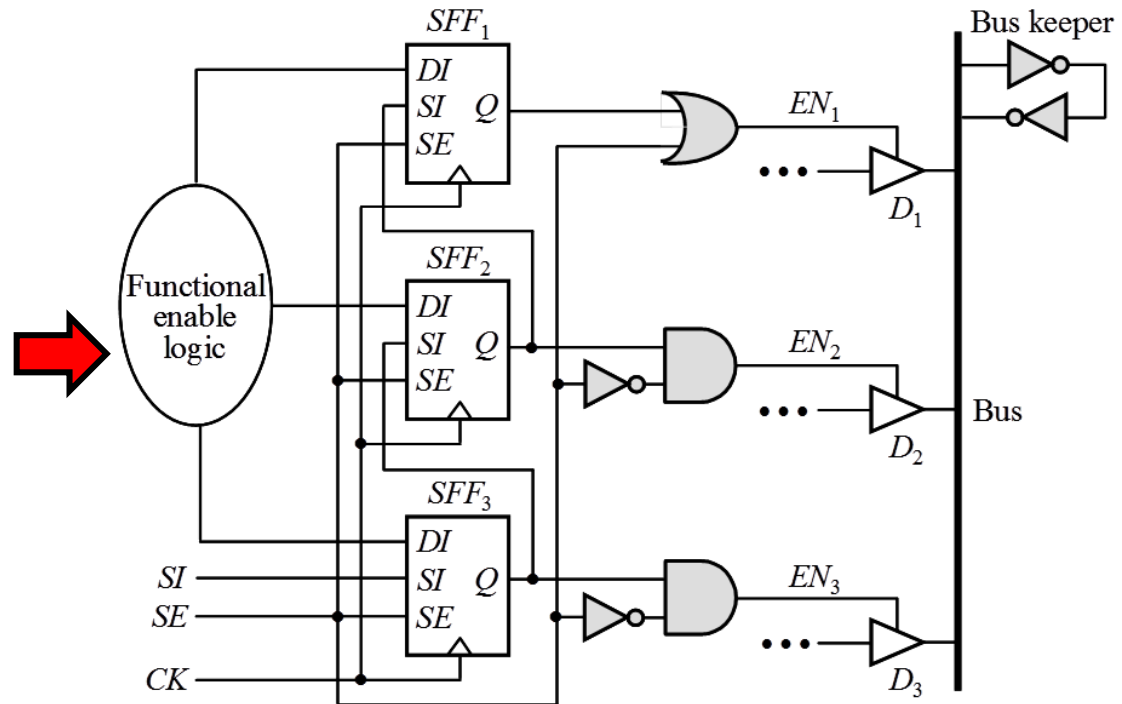
Successful Scan Design is Non-trivial

FFT

- Q: We make sure no contention when SE=1. How about SE=0?
 ♦ i.e. How to avoid contention during scan capture?



Original design



Testable design

No bus contention when SE=1
 No floating bus

FFT

- Q: What is wrong with this scan chain?
 - ♦ HINT: consider time skew

